MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATIONS, MARCH 2025
Third Semester

DSC-A - MG3DSCZGY200
PROTISTAN DIVERSITY AND ANIMAL DIVERSITY - NON CHORDATA – I

Duration: 1.5 hr Maximum Mark:50

PART A – MCQ (12 × 1 mark)Write all 12. (12 X 1 =12 marks)

1. Which of the following organisms is a pathogenic protist causing kala-azar in humans?
   a) *Schistosoma haematobium*
   b) *Leishmania donovani*
   c) *Wuchereriabancrofti*
   d) *Fasciola hepatica* [U][CO:1]

2. Which class of the Phylum Porifera includes the genus *Euplectella*?
   [K][CO:2]
   a) Demospongia
   b) Hexactinellida
   c) Calcarea
   d) Sclerospongia

3. Which of the following is a pathogenic nematode that causes filariasis in humans?
   a) Ancylostoma
   b) Enterobius
   c) Ascaris
   d) Wuchereriabancrofti [K][CO:4]

4. What type of canal system is found in *Sycon*?
   [U][CO:2]a)Asconoid
   b) Syconoid
   c) Leuconoid
   d) Rhagon

5. The class Turbellaria includes which of the following organisms? [U][CO:4]
   a) Fasciola
   b) Taenia
   c) Planaria
   d) Schistosoma

6. Which is the correct order of life stages in *Fasciola hepatica*? [U] [CO:4]
   a) Miracidium → Sporocyst → Redia → Cercaria → Metacercaria
   b) Sporocyst → Miracidium → Redia → Metacercaria → Cercaria
   c) Redia → Miracidium → Sporocyst → Cercaria → Metacercaria

d) None of the above

7. Which class does *Trypanosoma* belong to? [K] [CO:1]
   a) Ciliophora
   b) Kinetoplasta
   c) Sporozoa
   d) Rhizopoda

8. Which classification system introduced the Kingdom Protista? [K][CO:1]
   a) Two kingdom
   b) Five kingdom
   c) Three domain
   d) Eight kingdom

9. What is the role of choanocytes in sponges? [U][CO:2]
   a) Reproduction
   b) Defense
   c) Digestion and water flow
   d) Sensory reception

10. Which of the following is a member of phylum Ctenophora? [K][CO:3]
    a) Rhizostoma
    b) Planaria
    c) Pleurobrachia
    d) Trichinella

11. Which of these is NOT a plant-like protist? [U][CO:1]
    a) Euglena
    b) Volvox
    c) Paramecium
    d) Diatoms

12. Which of the following organisms shows metagenesis? [U][CO:2]
    a) Metridium
    b) Rhizostoma
    c) Obelia
    d) Hirudinaria

PART B – Short Answer (3 marks) Write any 6. (6 X3 =18 marks)

1. List the salient features of Phylum Ctenophora [K][CO:3]

2. Differentiate between diploblastic and triploblastic animals with examples.[U][CO:2]

3. Give a brief account of the canal system in sponges. [U][CO:2]

4. Mention any three parasitic Platyhelminthes affecting humans or dogs. [K][CO:4]
5. Explain the different types of body symmetry found in animals [U][CO:2]
6. Distinguish between Class Phasmidia and Aphasmidia of Nematoda with examples.[U][CO:4]
7. Describe the ecological adaptation in Chaetopterus. [U][CO:4]
8. Write a short note on the significance of Phylum Rhodophyta. [U][CO:1]

PART C – Short Essay (5 marks) Write any 4. (4 X5 =20 marks)

1. Write a brief note on the different parasitic nematodes affecting humans [U][CO:4]

2. Explain the life cycle of *Plasmodium* with emphasis on the human and mosquito hosts. [U][CO:1]

3. Write a short note on the polymorphism seen among Coelenterate [U][CO:3]
4. Compare the Plathyhelminth*Schistosoma* and*Taeniasolium* [U][CO:4]
5. Describe the process of conjugation in Paramecium. [U][CO:1]
6. Explain the basis of animal classification. [U][CO:2]

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